

FRIENDLY BEACHES, Australia

WMO: 949870

Lat: 41.995S

Lon: 148.279E

Elev: 56

StdP: 100.65

Time Zone: 10.00 (AUS)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	2.9	3.8	-2.0	3.2	8.6	-1.1	3.5	7.9	10.9	11.2	9.8	11.5	3.3	290	0.552

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	9.0	26.8	17.8	24.7	17.7	23.2	17.1	19.8	24.2	19.0	22.8	18.2	21.7	5.6	30

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.5	13.4	21.2	17.6	12.7	20.5	16.8	12.1	19.9	57.2	24.3	54.3	22.8	51.8	21.7	24.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
10.6	9.3	8.3	DB	0.7	33.9	0.8	3.1	0.2	36.2	-0.3	38.0	-0.7	39.8	-1.3	42.0	
			WB	0.2	21.8	0.6	0.9	-0.2	22.4	-0.6	23.0	-0.9	23.5	-1.3	24.1	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	13.6	17.7	17.5	16.5	14.1	11.9	10.1	9.7	10.1	11.6	12.9	14.5	16.2	(6)
(7)		DBStd	3.72	2.60	2.53	2.60	2.56	2.40	2.05	2.00	2.21	2.33	2.64	2.55	2.54	(7)
(8)		HDD10.0	102	0	0	0	1	8	24	30	27	9	3	1	0	(8)
(9)		HDD18.3	1818	42	41	69	131	198	247	267	256	202	171	119	76	(9)
(10)		CDD10.0	1398	239	211	202	123	68	27	21	29	57	93	134	193	(10)
(11)		CDD18.3	72	23	18	12	2	0	0	0	0	0	2	3	11	(11)
(12)		CDH23.3	384	128	83	49	9	0	0	0	0	1	23	32	59	(12)
(13)		CDH26.7	99	38	21	9	1	0	0	0	0	0	5	9	16	(13)
(14)	Wind	WSAvg	4.0	4.5	4.2	4.1	3.6	3.6	3.5	3.7	4.0	4.1	4.2	4.4	4.5	(14)
(15)	Precipitation	PrecAvg	734	62	45	55	67	48	49	58	52	59	64	83	(15)	
(16)		PrecMax	1260	242	194	239	377	336	194	186	208	166	206	209	337	(16)
(17)		PrecMin	359	0	0	0	0	0	3	0	7	0	0	3	0	(17)
(18)		PrecStd	228	60	43	48	65	60	40	48	46	43	44	47	78	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	30.9	29.6	27.5	24.3	19.9	16.8	16.5	18.6	21.3	26.2	27.4	28.9	(19)
(20)			MCWB	18.6	19.2	18.2	16.0	14.1	12.3	11.3	11.7	13.5	14.3	15.7	17.6	(20)
(21)		2%	DB	26.2	25.5	24.3	21.6	18.4	15.7	15.2	16.6	18.9	21.7	22.8	24.5	(21)
(22)			MCWB	18.4	18.9	18.2	15.9	13.4	11.8	10.7	11.2	12.3	14.0	15.6	17.0	(22)
(23)		5%	DB	24.0	23.6	22.6	19.9	17.2	14.8	14.2	15.2	17.3	19.4	20.7	22.6	(23)
(24)			MCWB	17.8	18.0	17.2	15.0	12.8	11.3	10.1	10.5	11.8	13.3	14.9	16.3	(24)
(25)		10%	DB	22.5	22.1	21.1	18.6	16.2	13.9	13.4	14.1	16.0	17.7	19.3	21.0	(25)
(26)			MCWB	16.8	17.0	16.4	14.4	12.2	10.5	9.7	9.9	11.1	12.4	14.2	15.6	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	21.1	20.8	20.6	18.2	15.7	14.5	12.7	13.1	14.8	16.6	18.2	19.6	(27)
(28)			MCDWB	25.8	25.2	24.9	21.2	17.2	15.2	14.5	15.5	18.7	20.8	23.2	25.6	(28)
(29)		2%	WB	19.7	19.7	19.2	17.0	14.5	12.7	11.6	12.1	13.5	15.2	16.8	18.3	(29)
(30)			MCDWB	24.3	23.9	22.5	19.8	17.0	14.9	13.9	15.0	17.0	19.1	21.2	22.6	(30)
(31)		5%	WB	18.7	18.9	18.3	16.0	13.5	11.6	10.9	11.3	12.7	14.2	15.9	17.3	(31)
(32)			MCDWB	22.9	22.2	21.2	18.9	16.4	14.1	13.4	14.3	16.0	18.0	19.7	21.1	(32)
(33)		10%	WB	17.7	18.0	17.3	15.0	12.7	10.8	10.2	10.5	11.8	13.3	14.9	16.3	(33)
(34)			MCDWB	21.6	21.0	20.2	18.0	15.7	13.4	12.9	13.6	15.2	16.8	18.3	19.9	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	9.0	8.5	8.2	7.9	6.9	6.7	6.8	7.4	8.4	8.8	8.5	8.8	(35)
(36)			MCDBR	11.6	10.7	9.8	9.7	7.8	7.4	7.4	8.4	9.7	11.1	11.0	11.1	(36)
(37)			MCWBR	6.2	5.9	5.7	6.0	5.3	5.0	4.9	5.3	5.9	6.1	6.0	6.0	(37)
(38)		5% WB	MCDBR	10.6	9.1	8.6	8.4	7.2	6.6	6.7	7.8	8.6	9.8	9.7	9.6	(38)
(39)			MCWBR	6.1	5.5	5.5	5.7	5.3	4.9	5.0	5.3	5.8	6.0	5.8	5.7	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.346	0.342	0.327	0.317	0.301	0.297	0.290	0.307	0.324	0.333	0.340	0.340	(40)	
(41)		taud	2.536	2.564	2.606	2.603	2.613	2.605	2.636	2.588	2.535	2.523	2.532	2.542	(41)	
(42)		Ebn at Noon	980	958	929	870	816	783	822	867	916	956	981	993	(42)	
(43)		Edh at Noon	107	99	87	75	62	57	60	74	91	102	107	108	(43)	
(44)	All-Sky Solar Radiation	RadAvg	6.61	5.56	4.38	2.99	1.96	1.55	1.78	2.60	3.90	5.16	5.95	6.77	(44)	
(45)		RadStd	0.46	0.26	0.20	0.19	0.12	0.12	0.09	0.17	0.26	0.26	0.44	0.38	(45)	

Historical Trends

		Heating		Cooling			Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	+0.28	N/A	N/A	N/A	+0.61	+0.59	N/A	N/A	N/A	N/A	(46)
(47)	Regional (27 neighbors)	+0.31	N/A	N/A	+0.75	N/A	N/A	N/A	-103	+110	+18	(47)

Nomenclature: See separate page